MILITARY SPECIFICATION

DETONATOR ASSEMBLY FOR FUZE, POINT DETONATING, M759

This specification is approved for use by the US Army Armament, Munitions and Chemical Center, and is available for use for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains the requirements and quality assurance provisions for fabrication of parts, assembly, and packing for a stab detonator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation-Packing, Methods of
MIL-R-398	-	RDX
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-650	-	Explosives: Sampling Inspection and Testing

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following Government documents drawings, and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

11834312	-	Fuze, Pd, M759
11834322	-	Cup, Detonator
11834323	-	Cup, Insert
11834324		Detonator Assembly
11834325	-	Mix, Primer
11834335	-	Spitback Lead Assembly
11834340	-	Detonator and Lead Propagation Test Assembly
11834345	-	Spitback Lead Assembly, Test

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the Contracting Officer.)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

2.2 Other publications. The following doucments form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, DC 20402. Orders should specify, “49 CFR100-199 (latest revision)”)

2.3 Order of precedence. In the event of a conflict between this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

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3.1 Materials. Materials shall be in accordance with applicable drawings and specifications (see 4.5.1).

3.2 Components and assemblies. The detonator shall comply with all requirements specified in Drawing 11834324, all asociated drawings, and with all requirements specified in applicable specifications and standards (see 4.5.1).

3.3 Moisture content.

3.3.1 Lead azide. The moisture content of the lead azide used in loading the detonator shall not exceed 0.15 percent when tested as specified in 4.5.2

3.3.2 Primer mixture. The moisture content of the primer mixture used in loading the detonator shall not exceed 0.15 percent when tested as specified in 4.5.3.

3.3.3 RDX. The moisture content of the RDX used in loading the detonator shall not exceed 0.15 percent when tested as specified in 4.4.5.

3.4 Functioning.

3.4.1 Sensitivity and output. The detonator shall function when initiated by a .004 inch diameter minimum flat firing pin tip struck with 1 1/4 in-oz energy on the first attempt and initiate the lead over an air gap of 0.200 inch minimum and cause the lead to indent a steel witness block to a depth of 0.025 inch or greater when tested as specified in 4.5.6.1.

3.4.2 Varicomp lead initiation. The detonator shall function on the first attempt and initiate the test lead over an air gap of 0.200 inch minimum and cause the lead to indent a steel witness block to a depth of not less than 0.025 inch when tested as specified in 4.5.6.2.

3.5 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be clean and free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample,as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Cup, Detonator	11834322	25
Cup, Insert	11834323	25
Detonator, Stab	11834324	233

4.3.2 Inspection to be performed. See MIL-A-48078 and Table 1 included herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Detonator Assembly and Metal Parts		SHEET 1 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, Detonator (Dwg. 11834322) Examination for defects	25		3.2	4.4.2.1
	Cup, Insert (Dwg. 11834323) Examination of defects	25		3.2	4.4.2.2
	Detonator Assembly (Dwg. 11834324)				
	Examination for defects	Note 1		3.2	4.4.2.5
	Functioning-sensitivity and output	200	Note 2	3.4.1	4.5.6.1
	FUnctioning-varicomp lead initiation	33	Note3,4	3.4.2	4.5.6.2
NOTE 1	The sample shall consist of the detonators to be functioned per 4.5.6.1 and 4.5.6.2.				
NOTE 2	Accept on one defect. Reject on two or more defects.				
NOTE 3	Demonstrate a 99.99% reliability at 95% confidence using the following accept/reject criteria: for n = 9, a = 0, r = 6 (minimum sample); n = 15, a = 1, r = 6; n = 20, a = 2, r = 6; n = 25, a = 3, r = 6; n = 29, a = 4, r = 6; n = 33, a = 5, r = 6 (maximum sample)				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Detonator Assembly and Metal Parts				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTE 4	Varicomp is a method for evaluation the interface between explosive components in which it can be determined by statistical analysis and testing that the reliability of a fuze explosive train can be predicted at high confidence levels with a small number of tests. This is done by varying the sensitivity of different explosives that are then substituted into the explosive train to determine the reliability limits under a penalty test situation. For reliability tests the acceptor design explosive is replaced by a less sensitive "varicomp" explosive. (Reference NWC TP 5789 12/75)				

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer, in accordance with the same specification and same specification revision, which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. RDX from not more than one lot.

d. Primer mixture consisting of batches blended by the same process. Each batch shall be composed of the same ingredients, each of which is to be from one interfix lot from one manufacturer.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup, Detonator	1 OF 1	11834322
			NEXT HIGHER ASSEMBLY 11834324

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Cup, split or cracked		0.40%	3.2	Visual
MINOR:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Cup, Insert		11834323
			NEXT HIGHER ASSEMBLY
			11834324

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Cup, split or cracked		0.40%	3.2	Visual
102.	Thickness through bottom of thin area		0.40%	3.2	Gage
MINOR:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 11834324
4.4.2.3	Detonator Assembly (Prior to Loading)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Moisture content of lead azide	Note1	Note5,6	3.3.1	4.5.2
102.	Moisture content of primer mix	Note2	Note5,6	3.3.2	4.5.3
103.	Moisture content of RDX	Note3	Note5,6	3.3.3	4.5.4
104.	Composition of primer mix	Note4		3.2	4.5.5
MINOR:	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.3	Detonator Assembly (Prior to Loading)				11834324
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Note 1	A sample of 2 grams shall be selected for each lot of lead azide at the time of loading.				
Note 2	A sample of 0.7 gram shall be selected for each lot of primer mixture at the time of loading.				
Note 3	A sample of 5 grams shall be selected for each lot of RDX at the time of loading.				
Note 4	A sample of 0.5 gram shall be selected for each lot of primer mixture at the time of loading.				

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Detonator Assembly (Prior to Loading)	3 OF 3	11834324
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Note 5	If the sample fails the test and loading has not begun, the lot of tested material shall be rejected until proper corrective action has been accomplished (where possible) as verified by reperforming the test. If loading has begun, action on the unloaded material shall be as specified above and all detonators loaded with the questionable material shall be rejected.				
Note 6	As an alternate this requirement can be met by drying the material in a 135°F ± 15°F oven for 72 hours. The oven is to maintain a relative humidity of less than 30%. Once dried the material must be maintained in a dry area for example using desiccant or oven.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11834324
4.4.2.4	Detonator Assembly (Charge Weight)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201.	Detonator cup lower charge weight, incorrect		Note 1	3.2	Balance or Qualified Column Height Inspection Station
202.	Detonator cup upper charge weight, incorrect		Note 1	3.2	"
203.	Insert cup lower charge weight, incorrect (Primer Mix)		Note 1	3.2	"

NOTES: 1. Check the first 5 loaded each shift and a minimum of 5 per hour thereafter if station is not qualified. No defects permitted. If a defect occurs, reject all product since last satisfactory check.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Detonator Assembly	1 OF 3	11834324
			NEXT HIGHER ASSEMBLY: See paragraph 6.1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1	Cup cracked or split		100%	3.2	Visual
2	Explosive exposed		100%	3.2	Visual
MAJOR:					
101.	Length		0.40%	3.2	Gage
102.	Diameter		0.40%	3.2	Gage
103.	Color incorrect or on wrong end		0.40%	3.2	Visual
104.	Functioning-sensitivity and output		0.40%	3.2	4.5.6.1
105.	Functioning-varicomp lead initiation	33	Note1,2	3.4.2	4.5.6.2
MINOR:	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70436(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.5	Detonator Assembly				11834324
					NEXT HIGHER ASSEMBLY
					See paragraph 6.1
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTE 1	Demonstrate a 99.99% reliability at 95% confidence using the following accept/reject criteria: for n = 9, a = 0, r = 6 (minimum sample) n = 15, a = 1, r = 6 n = 20, a = 2, r = 6 n = 25, a = 3, r = 6 n = 29, a = 4, r = 6 n = 33, a = 5, r = 6 (maximum sample)				
NOTE 2	Varicomp is a method for evaluation the interface between explosive components in which it can be determined by statistical analysis and testing that the reliability of a fuze explosive train can be predicted at high confidence levels with a small number of tests. This is done by varying the sensitivity of different explosives that are then substituted into the explosive train to determine the reliability limits under a penalty test situation. For reliability tests the acceptor design explosive is replaced by a less sensitive "varicomp" explosive. (Reference NWC TP 5789 12/75)				

NOTES:

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4.5 Test methods and procedures.

PRECAUTION - This specification covers sampling and testing of explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, Components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Moisture content of lead azide. The moisture content of the lead azide shall be determined at the time of loading. The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Mehtod 101.5 except, use a two gram sample,and cover the dish with a sheet of polytetrafluoroethylene.

4.5.3 Moisture content of primer mixture. The moisture content of the primer mixture shall be determined at the time of loading. A sample of 2.0 gram shall be selected. The sample shall be accurately weighed to the nearest milligram and dried to a constant weight of 60 degree C. plus or minus 5 degrees C., or in a desiccator containing anhydrous calcium chloride. The loss of weight shall be calculated as the percentage of moisture as follows:

$$\text{Percentage of Moisture} = \frac{100\ (A-B)}{A}$$

where:

A = The weight of the sample before drying.
B = The weight of the sample after drying.

4.5.4 Moisture content of RDX. The moisture content of the RDX shall be determined at the time of loading. The moisture content of the RDX shall be determined in accordance with MIL-STD-650, Method 101.5.

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4.5.5 Composition of primer mixture. The composition of th primer mixture shall be determined by performing the following tests in the sequence specified.

4.5.5.1 Barium nitrate. An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to tared medium-porosity, fritted glass crucible of 20 ml capacity. Three ml of lead azide saturated distilled water shall be added a temperature of 50 degrees C plus or minus 2 degrees C, agitate for one (1) minute, and, if necessary, lumps shall be broken up with a rubber policman attached to a glass rod. Suction shall b applied. This procedure shall be repeated five times, making a total of six, 3-ml extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighted. The loss in weigh shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Percentage of Barium Nitrate} = \frac{(A-B)\ 100}{W(1-.01M)}$$

where:

A = Weight of crucible and sample.
B = Weight of the crucible and sample after extractions with lead azide saturated water.
M = Moisture content of sample obtained from 4.5.3 in percer
W = Weight of sample.

4.5.5.2 Lead styphnate.

4.5.5.2.1 Basic lead styphnate (when applicable). The residue obtained in the determination of barium nitrate shall be extracted with six 5 ml portions of saturation ammonium acetate solution at about 25 degrees C, (hot saturated ammonium acetate shall not be used as it disolves antimony sulfide to some extent agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately two minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. Reserve the crucible and residue for the tetracene determination. The filtrate and washings shall be quantitatively transferred to a 2 ml calibrated volumetric flask, the beaker washed thoroughly wit distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml aliquot of this solution shall be transferred to 50 ml calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferre to acorex glass spectrophotometric cell having a width of 1

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centimeter (cm) and its absorbance determined at a wave length of 410 millimicrons. As reference, a similar cell containing saturated ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell the absorbance obtained as described above shall be corrected for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Percentage of Basic Lead Styphnate} = \frac{F-(A-B)}{W(1-.01M)D}$$

A = Absorbance of basic lead styphnate solution.
B = Absorbance of ammonium acetate solution.
D = Width of corex cell, cm.
M = Moisture content of sample obtained from 4.5.3 in percent.
W = Weight of sample.
F = Spectrophotometric factor. To determine the spectrophotometric factor (F) weight a 0.1 gm sample of pure lead styphnate quantitatively with 50 to 100 ml of diluted water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution, prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophotometric factor (F) for the lead styphnate.

4.5.5.2.2 Determination of normal lead styphnate (when applicable). Extract the residue obtained in the barium nitrate determination with sufficient 5 ml portions of ammonium acetate solution (saturated at about 25°C) until the washings are calorless. Agitate each portion for about one minute, then allow each portion to remain in contact with the residue for one additional minute. Finally, wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a

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second 100 ml calibrated volumetric flask and diluted to the mark with distilled water. Transfer a portion of this solution to a acorex glass spectrophotometric cell having a width of 1 centimeter (cm) and a similar cell containing saturated ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance obtained as the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution measure at a wave length of 410 millimicrons with absorbance of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Percentage of Basic Lead Styphnate} = \frac{F-(A-B)}{WD}$$

where:

A = Absorbance of basic lead styphnate solution.
B = Absorbance of ammonium acetate solution.
D = Width of corex cell, cm.
W = Weight of dried sample.
F = Spectrophotometric factor. To determine the spectrophotometric factor (F) weight a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of diluted water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution, prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophotometric factor (F) for the lead styphnate.

4.5.5.3 Tetracene. The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml beaker with a stream of water, 25 ml of water shall be added, and the contents of the beaker boiled on a hot plate for five minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and

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contents shall be dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Percentage of Antimony sulfide} = \frac{(D-E)}{W(1-.01M)}$$

where:

D = Weight of crucible and sample after treating with hot water.
E = Weight of crucible.
W = Weight of sample.
M = Moisture content of sample obtained in 4.5.3 in percent.

4.5.5.5 Lead azide. The lead azide content in the sample of a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, lead styphnate, tetracene and antimony sulfide.

4.5.6 Functioning.

4.5.6.1 Sensivitity and output. The sample detonators shall be randomly selected from each lot. The detonator shall be assembled with lead 11834335 in accordance with drawing 11834340. A weight of 0.25 ounce plus or minus 0.01 ounce shall be dropped on the firing pin, to function the detonator, from a height of five (5) inches maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. A detonator that fails to fire shall be classed defective. A detonator that fires but fails to initiate the lead sufficiently to produce a dent in the steel witness block of the required depth shall be classed defect if the detonator confinement sleeve expansion is 0.330 inch diameter or greater. If the confinement sleeve expansion is less than 0.330 inch diameter, the detonator shall not be classed defective, but the detonator classed defective and the result of a no-test. The indentation depth shall be measured using a probe with a 0.017 inch plus or minus 0.002 inch spherical radius tip.

4.5.6.2 Varicomp initiation. The sample detonators shall be randomly selected from each lot. The detonator shall be assembled with test lead 11834345 in accordance with drawing 11834340 and functioned as in 4.5.6.1. A detonator that fails as specified in 4.5.6.1 shall be classed defective. The indentation depth shall be measured using the probe specified in 4.5.6.1.

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4.5.7 Heat seal of sealed bag. The adequacy of the seal on the bag shall be determined in accordance with MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C. The detonator shall be packed in accordance With Department of Transportation Regulation 49 CFR100-199.

6. NOTES

6.1 Intended use. The detonator assemblies are intended for use in the following fuze -

30MM	M759	11834312

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit two (2) copies of designs as required to: AMCCOM, ATTN: DRSMC-QAF-S(D) , Dover, NJ 07801.

6.4 Check test for possible deterioration. The following test is designed to Prevent detonators that have deteriorated from being assembled into-major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert the contract provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

6.4.1 Sensitivity and output. A sample of two hundred and five (205) detonators randomly selected from each lot shall be tested as specified in 4.5.6.1.

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6.4.2 Varicomp lead initiation. A sample of thirty-three (33) detonators randomly selected from each lot shall be tested as specified in 4.5.6.2.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-A930)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-70436	DETONATOR ASSEMBLY FOR FUZE, POINT DETONATING , M759

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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